

BERT was the first model to show how bidirectionality can model performance NLP tasks. It has been used for various purposes, including improving the accuracy of Google's search results by allowing for a better understanding of the context and meaning of queries.

Other Large Language Models

Google's BERT and new advancements in the field inspired other companies to start building their own large language models. In the table below, we've listed some models developed prior to the truly groundbreaking GPT-3, released in June 2020.

Table 2 shows a timeline of consequential LLM releases.

Table 2. Large Language Model Release Timeline

Model	Company	Year	Brief Description
GPT-2	OpenAI	2019	A transformer-based language model designed to generate human-like text and perform various NLP tasks, including language translation, summarization, and question answering.
RoBERTa	Facebook	2019	A BERT-based model designed to improve the performance of NLP tasks by training the model on a larger dataset and using a more efficient training method.
DeBERTa	Microsoft	2020	A BERT-based model designed to improve the performance of NLP tasks by decoupling the encoder and decoder components of the model.
GPT-3	OpenAI	2020	An upgraded model of GPT-2 trained on a more massive dataset and capable of generating higher-quality outputs.

By [proving](#) that LLMs can be used for few-shot learning and excel without “large-scale task-specific data collection or model parameter updating,” GPT-3 would inspire companies to build even larger models, like Megatron-Turing Natural Language Generation with 530 billion parameters, PaLM with 540 billion, and WuDao 2.0 with impressive 1.75 trillion parameters.

Unsupervised and Self-Supervised Learning

BERT wasn't revolutionary just because it was a bidirectional model, but also because it was trained using *unsupervised learning*. Unsupervised learning refers to machine learning algorithms finding patterns in unlabeled datasets with no human intervention. In BERT's case, the model had to extract patterns from plain-language Wikipedia pages on its own during training. This is often considered to be AI in its purest form.

Unsupervised learning models use feedback loops to learn and improve their performance. This involves getting feedback on whether a prediction or classification was right or wrong, which the model uses to guide its future decisions.

Feedback loops are the reason why some differentiate between unsupervised and *self-supervised learning*. Self-supervised learning models don't have feedback loops but rather use supervisory signals

to get feedback during training. These signals are generated automatically from data without human annotation.

Both unsupervised and self-supervised learning techniques have one key advantage over supervised learning: they rely on the model to create labels and extract features on its own, rather than demanding human intervention. This helps companies train models without time-consuming data labeling processes or providing human feedback on the model's outputs.

Self-supervised learning is currently the dominant approach to pre-training large language models and is often recommended to enterprises that want to build their own.

Benefits of GPT Vs Bert

GPT (Generative Pre-trained Transformer) and BERT (Bidirectional Encoder Representations from Transformers) are both highly advanced and widely used natural language processing (NLP) models. However, they differ in their architectures and use cases.

GPT is a generative model that is trained to predict the next word in a sentence given the previous words. This pre-training enables GPT to generate coherent and fluent sentences from scratch, making it ideal for language generation tasks such as text completion, summarization, and question answering.

In distinction, BERT is a discriminative model that is trained to classify sentences or tokens into different categories such as sentiment analysis, named entity recognition, and text classification. It is a bidirectional model that considers both the left and right contexts of a sentence to understand the meaning of a word, making it highly effective for tasks such as sentiment analysis and question answering.

In terms of architecture, GPT uses a unidirectional transformer, whereas BERT uses a bidirectional transformer. This means that GPT can only consider the left context of a word when making predictions, while BERT considers both the left and right contexts.

Both GPT and BERT are powerful models that have revolutionized the field of NLP. Their choice depends on the specific task at hand, and researchers and practitioners often use a combination of both models to achieve optimal results.